

$$\frac{\mathbb{Z}[x, y]}{\langle 2, x^2, y^2 \rangle} \cong \frac{\frac{\mathbb{Z}[x, y]}{\langle 2 \rangle}}{\frac{\langle 2, x^2, y^2 \rangle}{\langle 2 \rangle}} \cong \frac{\mathbb{Z}_2[x, y]}{\langle x^2, y^2 \rangle}$$

$$= \{ a_0 + a_1 x + a_2 y + a_3 xy \mid a_i \in \mathbb{Z}_2 \}$$

Now,

$$\begin{aligned} & (a_0 + a_1 x + a_2 y + a_3 xy)^2 \\ &= a_0^2 + \underbrace{(a_1^2 x^2 + a_2^2 y^2 + a_3^2 x^2 y^2)}_{=0} + 2(\text{---}) \\ &= a_0 \end{aligned}$$

either $a_0 = 0$ or $a_0 = 1$